

MARGATE, South Africa

WMO: 685910

Lat: 30.859S

Lon: 30.343E

Elev: 505

StdP: 14.43

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	52.5	54.3	30.2	24.8	63.6	34.1	29.3	63.4	23.2	61.8	20.8	64.8	8.0	280	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.7	83.5	72.9	81.9	73.1	80.6	72.9	77.2	80.6	76.1	79.5	75.1	78.5	9.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	138.5	79.2	75.0	133.8	78.3	73.8	128.4	77.5	41.0	80.8	39.9	79.6	38.9	78.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.8	19.3	17.1	DB	48.6	89.5	6.9	2.4	43.6	91.2	39.6	92.6	35.7	94.0	30.7	95.7	
			WB	41.4	78.2	4.9	1.5	37.9	79.3	35.0	80.2	32.2	81.0	28.6	82.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	73.7	74.9	73.4	70.4	67.9	65.0	64.2	64.9	65.3	66.6	68.6	71.8
	DBStd	5.25	3.34	3.27	3.44	3.78	3.88	3.48	3.92	3.99	4.04	4.09	4.07	3.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	265	1	1	1	6	16	40	60	50	43	31	14	2
	CDD50	6883	735	696	726	611	555	450	440	462	459	515	557	677
	CDD65	1670	270	276	262	167	106	40	35	47	52	81	120	214
	CDH74	5442	1153	1367	997	366	209	80	87	109	76	102	226	670
	CDH80	424	79	114	65	23	29	10	12	15	12	11	12	42
(14) Wind	WSAvg	8.0	8.5	8.3	7.7	7.7	6.7	6.8	7.2	8.2	8.8	9.3	8.8	8.4
(15) Precipitation	PrecAvg	40.7	4.8	3.6	5.4	2.8	1.5	1.5	1.9	1.6	3.1	4.7	5.3	5.5
	PrecMax	65.4	13.9	8.4	13.0	7.0	4.7	13.4	10.4	5.7	11.0	11.3	13.9	12.4
	PrecMin	21.3	1.8	0.6	1.5	0.2	0.0	0.0	0.0	0.2	0.4	0.7	1.3	1.9
	PrecStd	10.6	2.5	1.9	3.4	1.9	1.3	2.8	2.6	1.4	2.2	2.5	3.0	2.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	85.1	83.7	82.6	83.5	80.8	81.3	82.4	81.4	80.3	81.7	83.9
		MCWB	76.0	75.6	74.9	71.9	64.8	60.6	60.3	62.0	65.9	69.3	73.0	74.8
	2%	DB	82.2	82.7	81.9	79.6	78.7	76.1	76.1	77.0	75.6	76.4	78.4	80.8
		MCWB	75.0	74.9	74.5	70.9	65.1	59.7	59.6	61.7	66.4	68.7	72.2	73.9
	5%	DB	80.6	81.2	80.3	77.7	76.0	73.2	72.6	73.5	73.0	74.4	76.5	79.0
		MCWB	74.1	74.4	73.7	70.2	64.7	60.0	59.3	62.1	65.7	67.8	70.9	72.9
	10%	DB	79.1	80.0	78.8	76.0	73.8	71.0	70.2	71.1	71.1	72.6	74.9	77.4
		MCWB	73.1	73.8	72.8	69.4	64.7	60.4	59.3	62.2	64.7	66.9	69.7	71.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.8	78.1	78.1	74.4	71.2	68.1	66.2	68.4	70.4	72.7	75.6	77.9
		MCDB	81.5	81.5	81.5	79.0	75.6	71.7	71.6	73.9	74.7	76.6	78.7	80.9
	2%	WB	76.8	76.8	76.4	72.8	69.3	65.7	64.5	66.7	68.6	70.4	73.7	75.7
		MCDB	80.2	80.5	79.8	77.2	73.6	70.2	69.7	71.6	72.6	73.9	76.7	78.6
	5%	WB	75.4	75.8	75.1	71.6	68.0	64.2	63.2	65.4	67.3	69.1	72.2	74.3
		MCDB	78.8	79.5	78.5	76.1	72.9	69.2	68.6	70.2	71.1	72.7	75.2	77.4
	10%	WB	74.2	74.8	73.9	70.5	66.8	62.7	61.9	64.1	66.0	68.0	70.8	73.0
		MCDB	77.9	78.6	77.6	74.8	71.7	67.7	67.5	68.9	69.8	71.6	74.0	76.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.7	8.8	9.9	11.0	12.0	11.8	10.7	9.6	9.1	8.6	8.2
		MCDBR	10.2	9.9	10.3	11.7	13.9	14.8	15.3	14.4	12.6	11.8	11.2	10.2
		MCWBR	5.9	5.3	5.9	7.1	9.0	10.2	10.5	9.3	7.9	7.2	7.0	6.0
	5% WB	MCDBR	9.4	9.3	9.5	11.0	11.7	12.0	12.0	11.4	10.9	10.2	9.6	9.5
		MCWBR	6.0	5.9	6.2	7.2	8.7	9.9	10.2	8.5	7.4	6.7	6.9	6.4
(40) Clear-Sky Solar Irradiance	taub	0.441	0.436	0.424	0.391	0.367	0.349	0.359	0.422	0.465	0.468	0.444	0.443	(40)
	taud	2.333	2.350	2.365	2.422	2.451	2.474	2.434	2.246	2.139	2.174	2.282	2.313	(41)
	Ebn at Noon	286	281	274	266	257	254	257	251	257	269	284	287	(42)
	Edh at Noon	43	41	38	32	28	26	28	38	47	49	45	44	(43)
(44) All-Sky Solar Radiation	RadAvg	1874	1809	1559	1278	1069	956	1022	1202	1400	1550	1724	1815	(44)
	RadStd	161	137	82	86	56	48	67	75	92	148	129	134	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page